

# William Caron-Bastarache

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## EDUCATION

### Université de Montréal

Montréal, QC

*B.Sc. in Computer Science (Expected 2027)*

2024 – Present

- Relevant coursework: Software Engineering, Algorithms & Data Structures, Operating Systems, Probability & Statistics, Linear Optimisation, Economics

### Cégep Édouard-Montpetit

Longueuil, QC

*College Diploma (Pre-University) in Business Administration with Mathematics*

2021 – 2024

## TECHNICAL SKILLS

**Programming:** Python, Java, SQL, Bash, C, HTML/CSS, JavaScript/TypeScript

**Data & ML:** PostgreSQL, Pandas, NumPy, scikit-learn, Matplotlib, psycopg

**Backend:** FastAPI, REST APIs, asyncio, Docker

**Frontend:** React, Next.js

**Testing & Quality:** pytest, JUnit, JaCoCo, PIT, GitHub Actions (CI/CD)

**Tools:** Git, GitHub, Linux, Make, Maven, Power BI, Claude Code, IntelliJ, VS Code, LaTeX

**Languages:** French (native) and English (fluent)

## PROJECTS

### Event-Driven Market Probability Engine (In Progress) | *PostgreSQL, Python, scikit-learn, Make, pytest* 2026

- Built a layered PostgreSQL data warehouse (raw/staging/analytics) and a Python + SQL pipeline covering 15 ETFs and 32,000+ daily rows, ingested concurrently with asyncio and rebuilt from a single Make command.
- Implemented walk-forward validation across 5 expanding folds with a 60-day purge/embargo, avoiding lookahead bias from random time-series splits; trained logistic-regression models and wrote predictions back to the warehouse.
- Enforced point-in-time correctness with 36 pytest cases and 6 database-level invariants that re-derive each stored value from its definition, with each invariant validated by deliberately breaking it.
- Backtested three positioning rules against an always-invested benchmark with transaction costs and reported the negative result: no strategy beat the benchmark (Sharpe 1.48 vs. 1.22 and 0.66).
- Codified the project's review discipline as Claude Code tooling: a CLAUDE.md file documenting architecture and invariants, PostToolUse hooks, and custom skills for data-leakage and documentation audits, with each finding converted into a permanent test.

### Routing & Mapping Software Quality Engineering (GraphHopper) | *Java, JUnit, JaCoCo, PIT* 2025

- Developed automated testing workflows for a graph-based routing and mapping engine.
- Measured test coverage before and after changes with JaCoCo and evaluated fault detection using mutation testing (PIT).
- Built a CI quality gate in GitHub Actions that fails builds on mutation-score regression using a persisted baseline.

### MaVille | *Java, Maven, REST, CLI App* 2025

- Designed and implemented a municipal management platform with data persistence, user profiles, notifications, and REST API integration.
- Implemented resident and contractor features, CSV/JSON persistence, and automated testing with JUnit.

## EXPERIENCE

### Incoming Data Scientist Intern – Optimisation

Winter 2027

*BRP*

*Québec, Canada*

- Incoming Data Scientist intern focused on optimisation, data analysis, and quantitative methods applied to operational problems.

### Lifeguard

2022 – Present

*City of Brossard*

*Brossard, QC*

- Ensure safety and respond to emergencies in high-traffic municipal pools.

## INTERESTS

Artificial Intelligence, Statistics, Finance, Software Development, Open-source Software, Fitness